TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Dr. Clemens KUBEIL	Email	Clemens.kubeil@ifam-dd.fraunhofer.de
Institute Name	Fraunhofer Institute for Manufacturing Technology and Advanced Materials, IFAM			
Address	Winterbergstrasse 28, 01277 Dresden, Germany			

2. Technology Overview

Title of Proposed Technology	*Large-Area Membrane-Electrode-Assembly (MEA) Materials and Manufacturing Technologies for Dynamic AEM Electrolysis*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		*Chemicals*	
	Medium Category	Small Category	Medium Category	Small Category
	New & Renewable Energy	*Hydrogen*	*Chemical Processes*	*Catalyst Application Technologies*
Concept of Proposed Technology	The proposed technology targets the development and optimization of material systems and manufacturing processes for high-performance Membrane Electrode Assemblies (MEAs) for anion exchange membrane water electrolysis (AEMEL). The project addresses the coupled design of electrodes (Catalyst Coated Substrates or Membranes CCS/CCM), porous transport layers (PTLs), and interfaces to AEM and related manufacturing processes to ensure efficiency, durability, and scalability of AEM electrolyzers. Key material functions include: • Tailored catalyst layer architectures for AEM-compatible electrodes, • Advanced PTL designs for optimized gas and liquid management, • Functional interlayers and coatings to improve electrode–membrane interfaces, • Systematic electrochemical validation under dynamic operation conditions representative of intermittent renewable energy input. • AST development for degradation and life-time improvements The proposed technology supports European and global hydrogen markets by enabling MEA			

Rationale for Joint R&D Project	designs compatible with industrial manufacturing constraints and by providing quantitative performance and durability benchmarks required for commercialization.
	Optimized MEA performance in AEM electrolysis cannot be achieved by isolated development of membranes, electrodes, or PTLs. Instead, integrated material and interface engineering, combined with application-oriented testing, is required.[1] Korean SME will link catalyst manufacturing expertise, scalable coating technologies, and system relevance, with Fraunhofer IFAM's surface engineering and electrochemical validation capabilities to translate innovations from lab to industrial relevant stacks for intermittent renewable energy operation. The joint approach reduces technical and commercialization risk by combining materials innovation with application-oriented validation.
Key Words	1. Membrane electrode assembly (MEA) 2. Electrode development (CCS/CCM) 3. AEM electrolysis 4. Porous transport layer 5. Accelerated stress test (AST)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	The objective of this project is to develop and validate material and process technologies for optimized MEAs for AEM electrolysis, integrating electrode layers, PTLs, and membrane interfaces into scalable and durable cell assemblies. The project aims to demonstrate a MEA construction (CCM/CCS) with stable performance under realistic and dynamic operating conditions, enabling maturation from laboratory-scale validation to prototype-level readiness. MEA architectures include catalyst-coated substrates and/or membranes, without restricting the approach to a single catalyst placement strategy. Specific objectives are: 1. Development of scalable electrode fabrication routes compatible with AEM materials. 2. Design and optimization of PTLs for enhanced gas bubble management and mass transport. 3. Functional coating of PTLs and/or membranes to improve interfacial contact and stability. 4. Electrochemical testing and durability validation under steady-state and dynamic operation. 5. Establishment of quantitative performance, durability, and scale-up benchmarks.			
R&D Target Specifications	The project targets the development of optimized membrane electrode assemblies (MEAs) in anion exchange membrane water electrolysis cells capable of operating at current densities of ≥1.0 A/cm² at a cell voltage of ≤1.9 V at ≤80 °C, under defined reference conditions. Durability targets include a voltage degradation rate of ≤100 μV/h and a maximum performance loss of ≤10% after defined accelerated stress testing and dynamic load cycling. Dynamic operation relevant to intermittent renewable energy input will be validated over a load range of 10–100% current density for at several >5000 cycles with faradaic efficiency >98%. Scalability of the developed materials and concepts will be demonstrated by producing MEA interface designs of ≥15 × 15 cm² active area, with a performance deviation of ≤10% between compared to small-area lab scale. Optimized PTL architectures and surface treatments are targeted to lower mass-transport-related overpotentials by ≥50 mV at current densities ≥2 A cm^{-2} compared to baseline systems. Successful achievement of these targets will demonstrate prototype-level maturity corresponding to TRL 5 for MEA material and process technologies suitable for further scale-up and system integration.			
R&D Plan	Work packages (WP) will be jointly executed by the Korean SME and Fraunhofer IFAM and consist of a phase I (pre-study, months 1-6) and phase II (Main R&D, months 7-36). **WP1 – MEA Material and Application Requirements (Months 1–6)** Definition of target MEA architectures (CCS, CCM and hybrid concepts), material interfaces, and operating conditions. Baseline AEM electrochemical testing and selection of electrode, PTL, and membrane reference systems, design choices and agreement on performance and durability			

	criteria. **WP2 – Electrode Fabrication and Functional Coatings (Months 4–24)** Development of scalable electrode fabrication routes (e.g. spray coating, blade coating, electrodeposition etc. in accordance with WP 1) and functional coatings on PTLs and/or membranes (CCS vs. CCM) including optimization of coating parameters, surface modification, catalyst layer microstructure and adhesion. Methods include microstructural analysis (Microscopy and SEM/EDX, XRD) and electrochemical cell and electrode testing. **WP3 – PTL Design and Gas Management Optimization (Months 10–24)** Tailoring of PTL structure, wettability, and surface properties to improve gas bubble detachment and mass transport in MEAs. Key tasks involve systematic variation of PTL structure and wettability combined with operando (Electrochemical Impedance Spectroscopy EIS) and ex-situ diagnostics (conductivity measurement, microscopy), leading to PTL designs with reduced mass-transport losses at high current density. **WP4 – Electrochemical Validation and AST (Months 16–30)** Performance and durability validation of selected MEA architectures under steady-state and dynamic operation. Methodologies include load cycling with defined test protocols, accelerated stress testing (AST), and post-mortem analysis, resulting in quantified degradation rates and lifetime trends applicable across MEA architectures. **WP5 – Large-Area MEA Prototypes and TRL 5 Demonstration (Months 24–36)** Transfer of optimized materials and processes to MEA prototypes (≥15 x 15 cm²) followed by validation under representative AEM electrolysis conditions. The work culminates in a TRL-5 demonstration, defined by performance, durability, and scalability verification of finalized MEA assemblies. **WP6 – Exploitation** WP6 addresses exploitation, IP positioning, and preparation of follow-up joint R&D or pilot-scale projects, as well as European market entry.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Fraunhofer IFAM will contribute with its expertise in applied research and validation of MEA material technologies, including: • Advanced electrochemical testing, AST, and degradation analysis, • Scale-up validation from small-area to large-area MEAs, • Quantitative benchmarking and technical roadmap development, • Electrode fabrication and functional coating development, • PTL design and gas management optimization. The institute will be responsible for MEA architecture development, electrochemical testing and validation at cell and short-stack level, as well as durability, and degradation mechanism analysis under realistic operating conditions. In addition, Fraunhofer IFAM will provide technical feedback for structural and process optimization, supporting the transition from laboratory-scale concepts to pilot-scale manufacturable PTL solutions.
Expected R&R of Korean SME Partner	The Korean SME will lead the development, optimization, and supply of catalyst and electrode coatings, specifically tailored for AEM electrolysis, with a focus on scalable electrode fabrication. This includes catalyst synthesis and formulation and coating procedures with iterative improvements based on electrochemical and analytical feedback. The SME will contribute to the MEA/electrode co-design, and translate the developments into relevant cell / stack designs. The SME will lead the assessment of manufacturability and cost aspects, ensuring that the developed materials and concepts are suitable for prototype validation and future commercialization. The Korean SME will lead scalable implementation and translation toward stack-relevant designs.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The project will deliver TRL 5 MEA material and process technologies for AEM electrolysis, including validated electrode layers, optimized PTLs, and improved membrane interfaces. The results will enable higher efficiency, improved durability, and scalable manufacturing of AEM electrolyzer components. From an industrial perspective, the project strengthens the competitiveness of Korean SMEs in European and in the global hydrogen market, supporting cost-effective and reliable green hydrogen production. The collaboration establishes a strong foundation for subsequent pilot-scale demonstration and commercialization and manifests Korea's position in the European and global hydrogen value chain. The project directly supports Korean SME entry into European AEM electrolysis supply chains

6. References

(1) Miller, Hamish Andrew, Karel Bouzek, Jaromir Hnat, Stefan Loos, Christian Immanuel Bernäcker, Thomas Weißgärber, Lars Röntzsch, and Jochen Meier-Haack. "Green Hydrogen from Anion Exchange Membrane Water Electrolysis: A Review of Recent Developments in Critical Materials and Operating Conditions." *Sustainable Energy & Fuels* 4, no. 5 (May 6, 2020): 2114–33. https://doi.org/10.1039/C9SE01240K.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization